

EECS 293
Software Craftsmanship
2015 Fall Semester

Quiz 1

Your name: _____

Your discussion section (time and location, or section number):

Check here if your discussion session is on Wednesday ☐

The test is 15 minutes, 10 points
Closed books and notes

1. (4 points) Briefly outline a possible scenario where you would choose a sequential over an iterative approach to software development.

2. (3 points) Briefly describe an issue that a software architecture should address.

3. (3 points) You have been promoted to software development manager and assigned to a brand new project. Your team has voiced widely varying opinions as to which language to use. Preferences range from mainstream languages like Java to specialized and esoteric languages. Briefly explain one factor that will affect the final decision.

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Quiz 2

Your name: _____

Your discussion leader: _____

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The test is 15 minutes, 10 points
Closed books and notes

1. (2 points) Explain and fix the problem in this code snippet:

```
if ((x != 0.) & (Math.sin(x) / x < MAX_VALUE))
```

2. A double variable called `federalDebt` stores a monetary value with the precision of one hundredth of a cent.
- a. (2 points) Briefly explain the problems with the current type (`double`) of `federalDebt`.

- b. (2 points) Suggest an alternative type for `federalDebt` and briefly explain your choice.

- 3. A method to write if-then-else statements involves covering the nominal case first and the error case later.
 - a. (2 points) Briefly explain the main idea of this method.

 - b. (2 points) Give an example where this method is insufficient to generate good code.

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Quiz 3

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The test is 15 minutes, 10 points
Closed books and notes

1. (8 points) Calculate the cyclomatic complexity of the following code snippets. Start the count at 0. Show your work.

a.

```
...  
if (a.equals(b) || a.testBit(0)) {  
    return a.gcd(MAX);  
}  
...
```

b.

```
...  
MAIN_LOOP:  
while (!a[i].equals(b[i])) {  
    if (a.gcd(MAX) > MIN) {  
        break MAIN_LOOP;  
    } else {  
        b[i] = MIN;  
    }  
}  
...
```

c.

```
...
switch (product) {
case 0WATCH:
    sum = 0
    for (i = 0; i < n; i++)
        sum += a[i];
    break;

case 0TV:
    sum = a[0] > MIN ? b[0] : b[1];
    break;

default:
    sum = 0;
}
```

2. (2 points) Rewrite the following code snippet to eliminate the repeated invocation of `fetchX` with no increase in complexity.

```
...
x[0] = fetchX();
for (int i = 1; i <= n; i++) {
    y[i] = trigFunction(x[i-1]);
    x[i] = fetchX() + y[i];
}
...
```

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Quiz 4

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The test is 10 minutes, 10 points
Closed books and notes

1. (3 points) Briefly explain the purpose of packaging Operating System dependencies in a separate subsystem.

2. (4 points) Give an example of semantic coupling involving an object initialized to an invalid state.

3. (3 points) Explain the benefits of making central points of control and give an example.

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Quiz 5

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The test is 10 minutes, 10 points
Closed books and notes

1. (7 points) Find an appropriate name for each routine given the following short descriptions. (In the following, “this” refers to the object to which the method belongs.)
 - a. Return the name of this employee.
 - b. Add to this employee the W-2 form for the current fiscal year, and return whether the employee was modified by this operation.
 - c. Find out and return whether this employee qualifies for Medicare.
 - d. A static method of AbstractEmployee to return a new employee of the specified type.
 - e. Convert a distance from kilometers to light years, and return it.

- f. Returns an estimated size of this relation in bytes.
- g. Returns the name associated with a leaf in this tree.

2. (2 points) Define the concept of *functional cohesiveness*.

3. (1 point) Give an example of a procedure that takes a single modify parameter. Provide the full method declaration, consisting of both method header and body.

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Quiz 6

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The test is 10 minutes, 10 points
Closed books and notes

1. (3 points) Suppose that a piece of pseudo-code failed to specify whether a collection should be sorted in non-decreasing or in non-increasing order. Give an example of a problem that may arise later on in the development cycle.
2. (4 points) Briefly explain the benefits of tracing pseudo-code execution.

3. (3 points) Calculate the cyclomatic complexity of the following code snippet. Start the count at 0. Show your work.

```
while (i < n && i == (a[i] > 0. ? a[i] / b[i] : -a[i])) {  
    if (i == b[i]) i++;  
    i++;  
}
```

Quiz 7

Check here if your discussion session is on Wednesday ☐

```
void methodUnderTest(Double x, Double y, Double z) {  
    Double w = 0.;  
    if (x < 0.) {  
        w = y < z ? Math.sqrt(z-y) : Math.sqrt(-x);  
    }  
    return w;  
}
```

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Quiz 8

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The test is 10 minutes, 10 points
Closed books and notes

1. (4 points) Briefly explain the problems with the following code snippet.

```
if (x == null) {  
    throw new Exception("x should not be null")  
}
```

2. (3 points) Briefly explain the concept of a *barricade* in defensive programming.

3. (3 points) Briefly explain the *singleton* design pattern and give an example of a standard singleton in the Java programming language.

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Quiz 9

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The test is 15 minutes, 10 points
Closed books and notes

1. (4 points) Suppose that a class `Builder` has a method called `addCompatible` that returns this `Builder`. Argue whether or not the instruction

```
builder = (new Builder()).addCompatible(s1).addCompatible(s2)
```

violates the Law of Demeter.

2. (6 points) Give an example of a defensive copy that is needed to ensure that a class is immutable. Explain your reasoning.

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Quiz 10

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The test is 15 minutes, 10 points
Closed books and notes

1. (6 points) Give an example in Java of a class self-use of overridable methods, including the relevant documentation. (Minor syntax errors are unimportant.)

2. (4 points) Briefly define the concept of *skeletal implementation*.

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Quiz 11

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Your discussion leader: _____

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The test is 15 minutes, 10 points
Closed books and notes

1. (4 points) Calculate the cyclomatic complexity of the following code snippets. Start the count at 0. Show your work.

```
...
while (!a.equals(b)) {
    switch (a) {
        case 0:
            return a.gcd(MAX);
            break;
        case 1:
            a--;
            break;
        default:
            b++;
    }
    ...
}
```

2. (2 points) Briefly describe a problem that arises if commenting are not integrated in the development cycle.

3. (2 points) Explain the problem(s) with the following comment:
- ```
/* Use right shift to divide by two. Substituting the right
 shift operation cuts the loop time by 75% */
... 50 lines of code ...
element = element >> 1;
```

4. (2 points) Give a short example of a comment used to differentiate between the input and output arguments of a method.